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The Implementation of Next-Generation PDC Bits Featuring Articulating Gauge Pads and Integrated Vibration Sensors Marks a Significant Step Forward in Advancing Digital Drilling Practices

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Directional flexibility in drill bit design plays a critical role in modern drilling operations, offering significant advantages in reducing well construction time. One of the primary challenges is achieving efficient rock failure while maintaining accurate borehole trajectory - especially in build-up and tangent sections.

Bit steerability is governed by multiple interrelated factors, including cutting structure, bit profile, presence of vibration mitigation features, gauge configuration design and length, lateral aggressiveness, and depth of cut (DOC), as discussed by [Barton et al. \(2007\)](#), [Chen et al. \(2007\)](#), [Rebrikov et al \(2019\)](#) and others. The primary objective of a PDC drill bit is to maximize rate of penetration (ROP) and total drilled footage, while adhering to the planned well trajectory. Achieving this goal requires careful control of operational parameters such as weight on bit (WOB), rotary speed (RPM), and drilling fluid flow rate. These must remain within optimal ranges to improve ROP while minimizing vibration-induced damage and other drilling dysfunctions. Directional drilling in build-up sections is often constrained by the mechanical limitations of positive displacement motors (PDMs), particularly when high bend angles restrict allowable RPM. Although under-gauge bits can improve build-up rates (BUR), they could sacrifice degrade borehole quality.

This paper introduces a next-generation PDC bit featuring an articulating gauge pad that dynamically transitions between different under-gauge profiles while drilling. Integrated vibration sensors at the bit enable high-resolution downhole data acquisition, supporting detailed post-run analysis of dynamic behaviour and performance.

Introduction

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This paper introduces a next-generation PDC bit featuring an articulating gauge pad that dynamically transitions between different under-gauge profiles while drilling. Integrated vibration sensors at the bit enable high-resolution downhole data acquisition, supporting detailed post-run analysis of dynamic behaviour and performance.

Working Principle of new articulating gauge pad PDC bit

The Articulating Gauge Pad feature ([Fig. 1](#)) consists of movable gauge pad elements that actively adjust the bit's lateral depth of cut (DOC) and side cutting efficiency (SCE) during drilling. This dynamic adaptation enables the bit to respond effectively to changing downhole conditions, particularly in directional drilling sections where high steerability is required.



Figure 1—Articulating gauge pad design feature

The primary objective of this design is to combine the directional responsiveness of short gauge pads with the stabilizing advantages of longer gauge pads. Shorter gauge pads tend to increase build-up rate (BUR) and directional responsiveness, but they may lead to greater vibration and poorer borehole smoothness. Conversely, longer gauge pads improve borehole quality and reduce tortuosity but limit steerability ([Mensa-Wilmot et al. 2006](#)). By allowing real-time transitions between these two profiles, the articulating gauge pad offers an optimal balance between steerability and stability.

The internal mechanism is a simple, robust design consisting of a spring stack, piston, retention pin, and a diamond-coated domed PDC element. The system allows the gauge pad profile to adjust in response to side forces encountered during drilling. When the lateral force from the formation exceeds the pre-calibrated spring resistance, the piston compresses, causing the domed PDC element to retract inward. This temporarily

reduces the effective gauge diameter, allowing the bit to cut more aggressively on the side - enhancing build-up rate when needed.

Once lateral forces diminish, the spring stack returns the piston to its original position, restoring the full-gauge profile. This passive, self-adjusting mechanism allows the bit to adapt continuously to formation changes - without the need for surface control or downhole actuation.

The articulating gauge pad technology uses customized pistons with precise lateral movement, tuned specifically for directional or tangent sections. This configuration optimizes side-cutting aggressiveness when steering is required and enhances lateral stability when holding angle. Spring stack force calibration has been refined using advanced in-bit sensor measurements systems showed on Fig. 2, allowing for tailored performance across diverse lithologies and trajectories.



Figure 2—In-Bit sensors measured system schematic.

Bending-on-Bit (BOB) Load Transfer and Moment Arm Estimation

Compressive force to fully push tacker into housing could be calculated as shown in Eq. 1.:

$$F = K * d \quad (1)$$

Where F - Side force applied to articulating gauge piston; K- Spring constant (depends on number of stack); d - piston standout height (Fig 3).

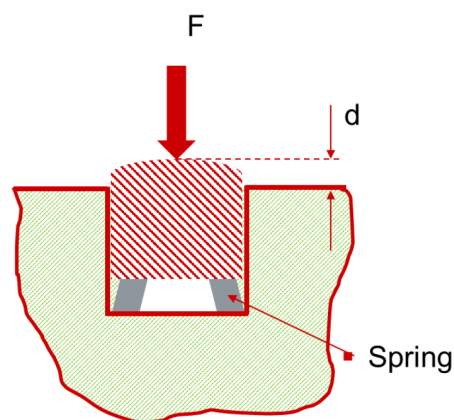


Figure 3—Compressive force acting on articulating gauge pad piston during formation contact.

The bending-on-bit (BOB) force acting on the articulating gauge pad can be estimated by scaling the BOB measured at the bit shank (centerline) to the radial trajectory of the articulating gauge pad piston. This approach assumes a linear transfer of the bending moment across the bit body, allowing force scaling based on relative radial distances.

The side force on articulating gauge pad (Fig 4) could be estimated from the BOB using this equation

$$F = \frac{BOB}{Moment\ Arm} \quad (2)$$

Where BOB - is obtained by scaling the BOB measured by an in-bit sensor to the piston axis center; Moment Arm – measured from the extreme of cutter profile to articulating gauge part. This dynamic load affects cutter contact distribution and articulating gauge pad activation during directional drilling.

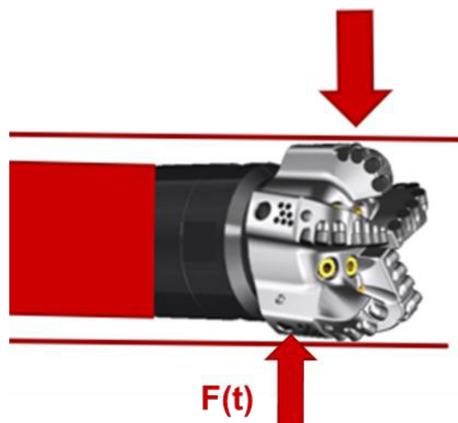


Figure 4—Illustration of bending-on-bit (BOB): force $F(t)$ applied to the drill bit due to combined axial and lateral constraints within the wellbore.

The effective moment arm is defined as the distance from the extreme point of the cutter profile to the centre line of the articulating gauge piston trajectory, as illustrated in Fig. 5. This moment arm governs the mechanical advantage by which BOB loads are transmitted to the articulating gauge pad, influencing the trigger threshold of pad retraction and side-cutting response.

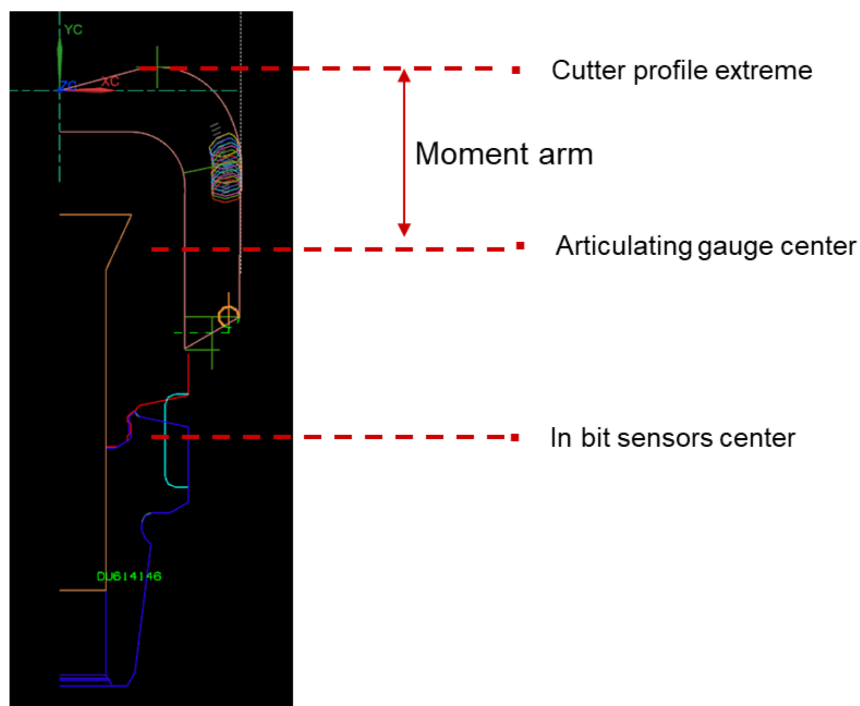


Figure 5—Cross-sectional view of articulating gauge pad PDC bit.

Using in-bit measured data by advanced sensor system, the Bending-on-Bit is collected for an offset bit run in the target application (Fig 6).

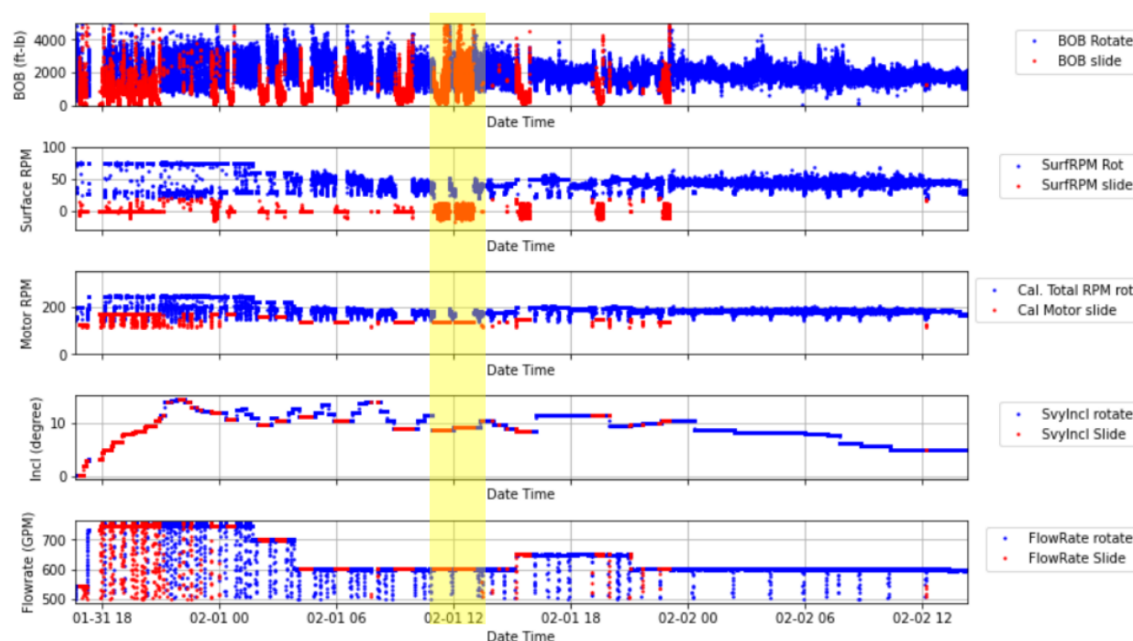


Figure 6—Advanced in Bit sensor Data for Offset Well

The sensor dataset is reviewed and divided into sliding and rotating mode (Fig 7). From this dataset, the average Bend-on-Bit is calculated in sliding mode. In the example shown in Fig 7, the average Bending-on-Bit is 1,750 lbf. This moment is then translated into the desired spring force by dividing the BOB value by the moment arm of the bit (see Equation 2). Thus, in-bit sensor data is used to tailor piston spring force by using available offset data to set the spring force to collapse the spring assembly in rotating mode, enabling steerability, and engage in sliding mode to improve stability.

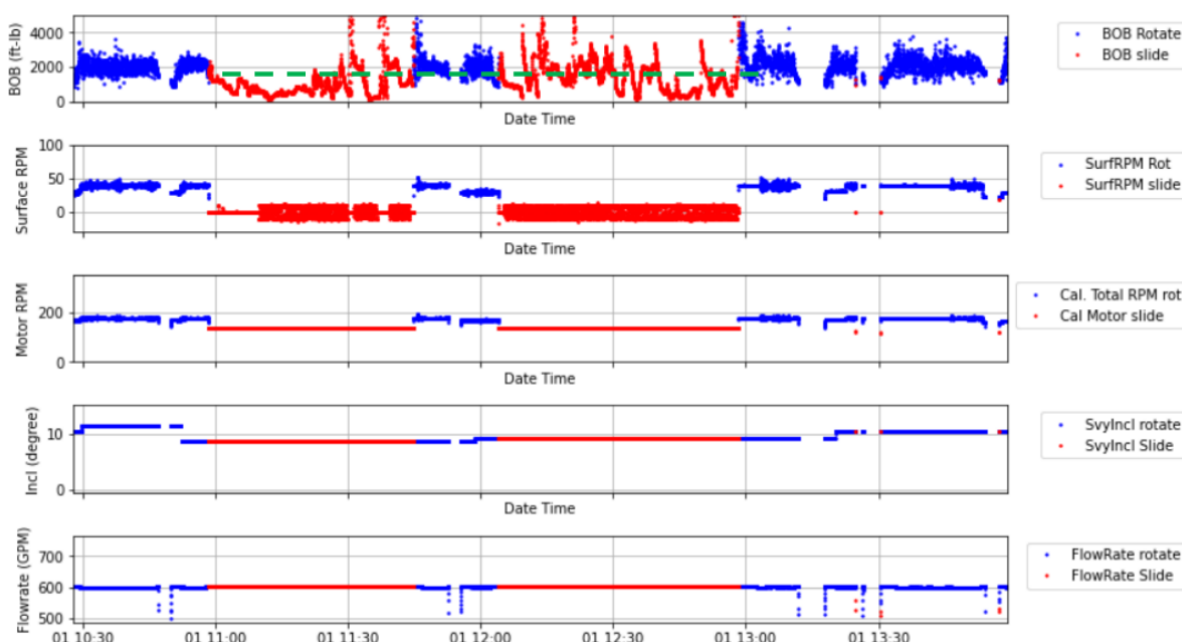


Figure 7—Zoomed in Section of In-bit sensor Data for Offset Well

This innovative articulating gauge system has demonstrated multiple benefits in field trials, including:

- Reduced vibrations
 - Springs provide vibration dampening
- Improved borehole quality (lower tortuosity)
 - Improved frictional interface from smooth diamond gauge pad feature
- Enhanced directional control in curve section
- Increased drilling efficiency

All of these advantages have been achieved without compromising rate of penetration, as validated by field results across various basins and drilling programs.

Field test results

8.5" field trial

One of the first field trials of the articulating gauge pad technology, documented by [Rebrikov et al. \(2021\)](#), was conducted using a bit equipped with in-bit vibration sensors only. The test was carried out in a curve section using an 8.5" PDC bit with the articulating gauge feature, as shown in [Fig. 8](#).



Figure 8—8.5" PDC drill bit with articulating gauge part.

During oriented drilling (slide mode), no significant reactive torque fluctuations were recorded, even with a consistent weight on bit (WOB) maintained across the entire interval. This enabled high rates of penetration (ROP) in both slide and rotary modes, indicating improved steerability and stability compared to fixed-cutter (FC) bits previously used in the same project (see [Fig. 9](#)). The adaptive gauge pad configuration significantly enhanced directional control, reduced the need for trajectory corrections, and enabled reaching total depth (TD) in a single run.

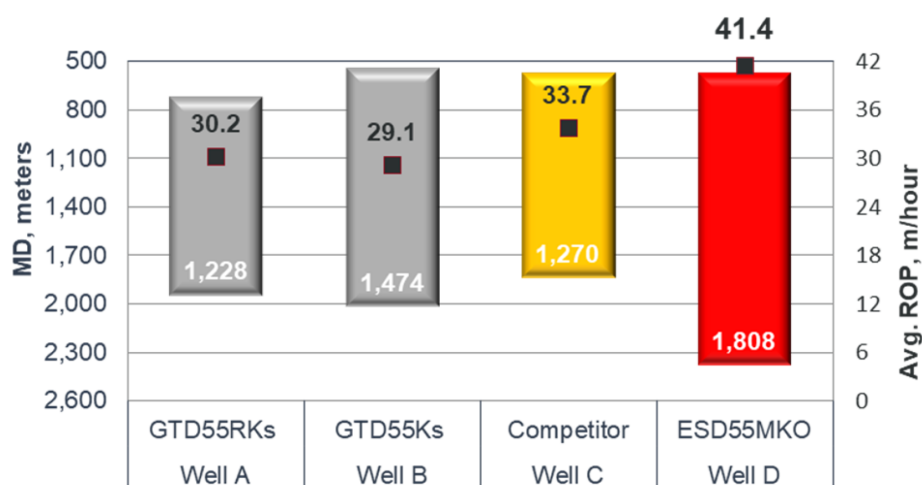


Figure 9—Offset wells comparison.

The trial results demonstrated an average ROP improvement of 18% compared to the best offset performance, which had been achieved using a conventional PDC bit in a less complex directional application. A total of 1,808 meters were drilled with minimal dysfunction, recorded at just 7%, based on data from in-bit vibration sensors (Fig. 10). This was corroborated by a favorable bit dull grading of 0-1-CT-C,S-X-I-NO-TD, indicating minimal wear and no major damage (Fig. 11). The signal processing and interpretation techniques applied to these sensor measurements were previously detailed by [Rebrikov et al. \(2021, SPE-206437-MS\)](#). These results validate the effectiveness of the articulating gauge pad in facilitating smooth and efficient drilling, ultimately contributing to improved well construction performance.

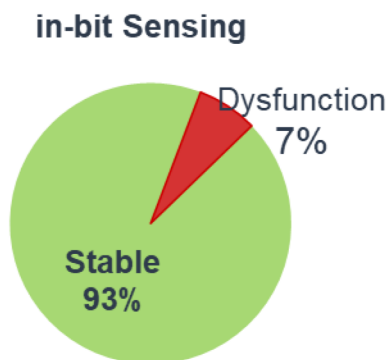


Figure 10—Post-run in-bit vibration sensing summary.



Figure 11—Post-run bit dull condition.

The implementation of this design feature marked a significant step forward in drilling efficiency, delivering measurable improvements in both bit performance and overall well construction outcomes in the 8.5" section. Incorporating BOB measurements will enable a more comprehensive understanding of bit–formation interaction and further optimization of articulating gauge pad behaviour under varying downhole loading conditions.

12.25" Field trial Midland Basin – Shallow Intermediate Deployment with advanced at bit measurement sensors

To better understand and fine-tune the spring force response of the articulating gauge pad, advanced in-bit sensor modules were incorporated to measure downhole weight on bit (DWOB), downhole torque on bit (DTOB) and downhole BOB. These parameters are critical for identifying the force thresholds at which the piston retracts and the gauge transitions from full-gauge to under-gauge configuration.

The methodology employed in this study expands on previously validated applications of in-bit sensor technology in challenging drilling environments. Notably, [Witt-Doerring et al. \(2023\)](#) demonstrated how embedded sensors can detect drilling dysfunctions such as ineffective weight transfer, string buckling, and lateral vibrations - highlighting the advantages of memory-based measurements and direct load monitoring at the bit.

During the run, the sensor data revealed several valuable operational insights, including:

- Analysis of DWOB and SWOB across various intervals to evaluate weight transfer effectiveness;
- Correlation between surface torque and downhole torque to understand torsional loading behaviour;
- Identification of backward whirl frequencies using spectrogram analysis.

Complementing the bit-mounted sensors, MWD tools located along the BHA collected additional string-level vibration data, enabling cross-comparison and alignment of dynamic responses between the bit and the drillstring.

In this trial, strain-based and motion based sensors were integrated within the bit shank and operated in memory mode at a high-frequency sampling rate of 1024 Hz. This setup allowed for direct, high-resolution acquisition of WOB, TOB, BOB and multi-axis dynamic responses at the cutting structure, providing valuable insight into energy transfer, bit dysfunctions, and overall drilling performance.

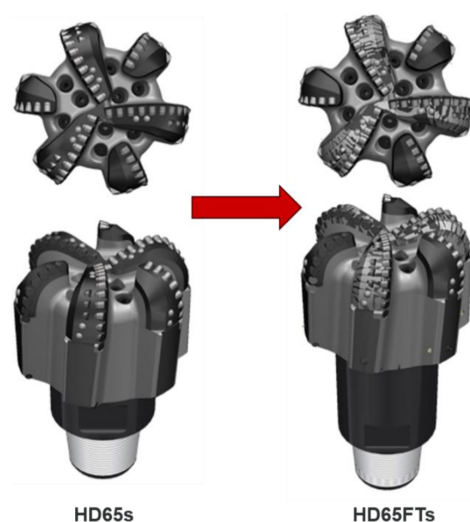
The system captured multiple vibration modes - including axial, lateral, and torsional - at the bit–rock interface. These data were critical for diagnosing the bit's mechanical behavior in real time. [Figure 3](#) presents a schematic of the sensor system integrated into the bit shank.

As summarized in [Table 1](#), the sensor suite includes vibration, shock, RPM, magnetometer, and strain measurement channels. These measurements enable post-run analysis of dysfunction severity and allow for the calculation of at-bit Mechanical Specific Energy (MSE). Comparing this downhole MSE with conventional surface-calculated MSE provides a more accurate understanding of energy transfer efficiency and mechanical losses in the BHA. This insight is essential for optimizing drilling parameters and improving bit design.

Table 1—Advance In-Bit Sensor Specifications

VIBRATION	Axes	3
	Range	+/- 8g
SHOCK	Axes	3
	Range	+/- 200 g
MAGNETOMETER	Axes	3
	Range	+/- 16 gauss
ACCEL/MAG RPM	Axes	3
	Range	0 to 1200 rpm
GYRO RPM	Axes	3
	Range	+/- 666 rpm
Strain	Axes	WOB/TOB/BOB

The HD65FTS bit with articulating gauge pad and integrated sensor technology was deployed in the shallow intermediate section of wells in Midland County, Texas, targeting formations between 1,700 and 7,400 ft. All wells were drilled using an 8" motor BHA with a 7/8" motor (0.16 rev/gal, 1.5°/1.83° bend), designed to build inclination from 0° to 4-24°, then hold to total depth (TD). This new bit shared the same cutting structure as the reference HD65S bit, with the only design difference being the articulating gauge pad system (Fig. 12).

**Figure 12—12.25" Bit design changes.**

This graduated offset design enabled a smooth transition in lateral aggressiveness, allowing the bit to adjust dynamically between high steerability during build sections and increased stability in tangent intervals. The PDC-topped piston system at the core of the articulating gauge pad design responded to real-time lateral loads by fully contracting during aggressive sliding and partially retracting in rotation, without requiring active surface control.

First field performance data showed: Consistent ROP, controlled build rates, reduced sliding time and lower borehole tortuosity. Collectively, these results contributed to the fastest well drilled in the test series (Fig. 13).

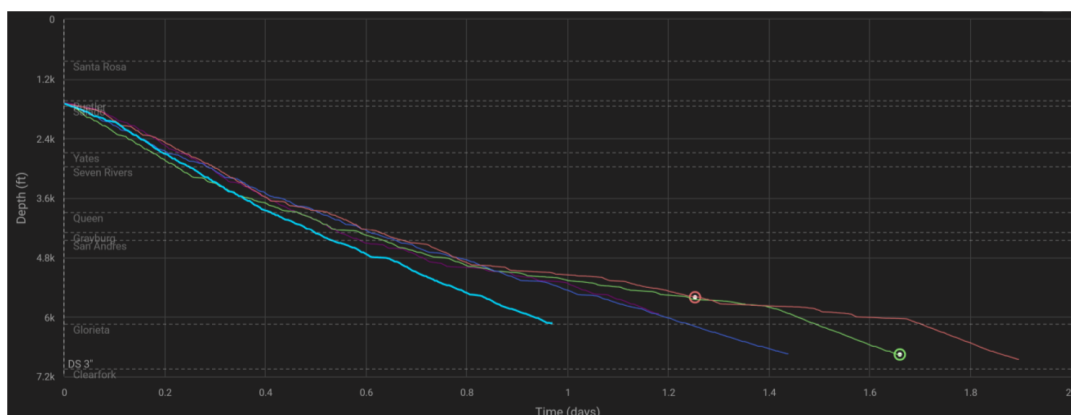


Figure 13—Depth vs. Time comparison for multiple wells using articulating gauge pad PDC bits in Midland Basin.

After the run, high-frequency data from the in-bit sensors were retrieved and down sampled to 1 Hz to align with surface-acquired parameters for depth-time correlation. This processing approach followed the method described by Kouzaiha et al. (2023, SPE-215976-MS), in which high-resolution in-bit data were used to evaluate dysfunctions and energy transfer in lateral drilling.

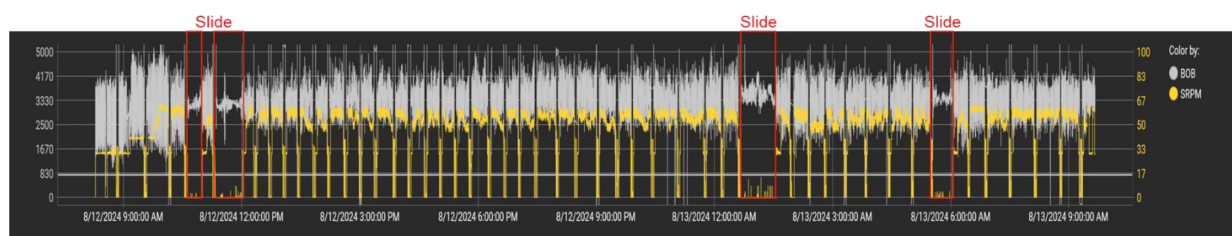


Figure 14—BOB measured from integrated sensor and surface RPM readings

The bend-on-bit (BOB) measurements recorded from the integrated sensor technology demonstrate that the bit side forces are higher while rotating vs sliding with bent housing motor assembly.



Figure 15—Dull photos of bit used in trial show the cutting structure to be in good condition with minimal wear. There was some wear observed on the gauge elements indicating they were engaged while drilling.

The HD65FTS bit was subsequently deployed in multiple wells and consistently outperformed offset runs (Fig. 16).

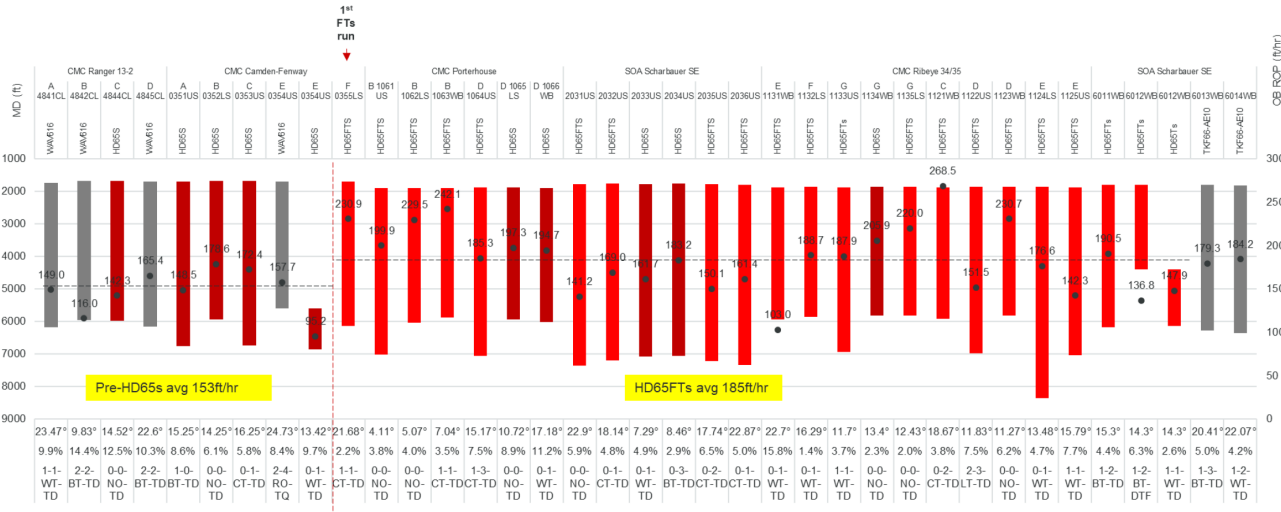


Figure 16—Intermediate section Midland Country Offset wells performance comparison.

Performance data for different bit types were grouped by inclination range and summarized in Table 2:

- 0–10° Inclination: HD65FTs demonstrated higher ROP and reduced slide percentage compared to both the reference HD65 bit (fixed gauge pad) and the other PDC616 design. This improvement reflects enhanced steerability and better toolface control, especially in early build sections.
- 11–15° Inclination: HD65FTs continued to show faster drilling rates and lower sliding requirements relative to the HD65 reference bit, highlighting its effectiveness in steeper build angles where gauge pad retraction is more frequently activated.

Table 2—Performance comparison by inclination range

Inclination Max (°)	Bit Type	Count	Avg Ftg (ft)	Avg OB ROP (ft/hr)	Avg Slide ROP (ft/hr)	Avg Slide %
0–10°	HD65FTS	3	4415	219.85	75.14	3.8%
	HD65S	3	4885	178.14	58.72	5.5%
	PDC616	1	4273	115.99	50.11	14.4%
11–15°	HD65FTS	3	4415	219.85	75.14	3.8%
	HD65S	3	4885	178.14	58.72	5.5%
	PDC616	1	4273	115.99	50.11	14.4%

These results support the conclusion that the articulating gauge system offers superior directional performance and drilling efficiency across a range of build rates compared to conventional gauge pad configuration. This confirmed that the articulating gauge pad mechanism effectively adapts to directional drilling demands, improves borehole quality, and contributes to greater drilling efficiency, especially in curved and transitional sections.

Conclusion

The articulating gauge pad system, when combined with advanced in-bit sensor technology, demonstrated measurable performance improvements in different applications. Compared to conventional gauge pad configurations, this unique design:

- Improved directional performance and ROP

- Reduced sliding time
- Maintained borehole quality
- Lowered dysfunctions
- Enhanced bit durability

These results confirm the articulating gauge pad's ability to adapt dynamically to drilling demands and its potential to significantly enhance drilling efficiency in curved and transitional intervals.

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